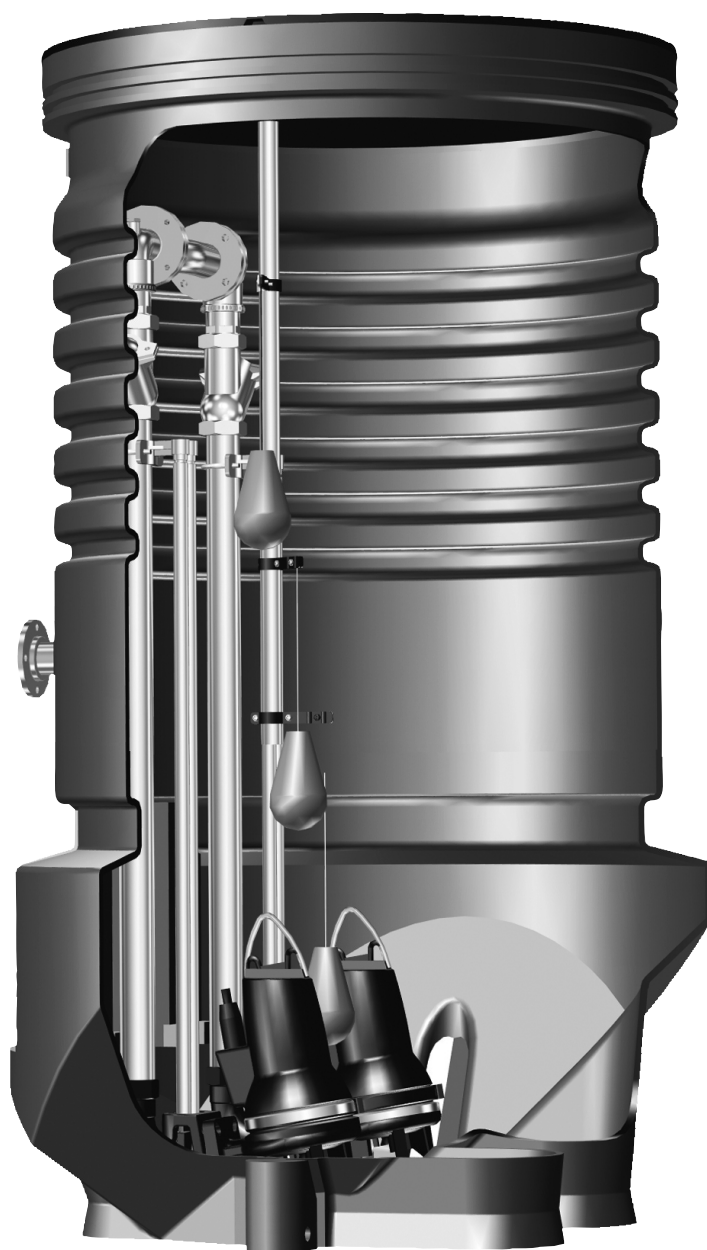


Pumping station systems

PS.R.05-17

Installation and operating instructions



Pumping station systems

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English (GB) Installation and operating instructions

Original installation and operating instructions

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1. General information



Read this document before you install the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1.1 Hazard statements

The symbols and hazard statements below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.



DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.



WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.



CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The hazard statements are structured in the following way:

SIGNAL WORD

Description of the hazard

Consequence of ignoring the warning

- Action to avoid the hazard.

1.2 Notes

The symbols and notes below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.



Observe these instructions for explosion-proof products.



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.



Tips and advice that make the work easier.

2. Safety instructions

Transporting the product



DANGER

Crushing hazard

Death or serious personal injury

- Place the pit on even ground to prevent the pit from overturning.

Precautions during lifting



DANGER

Crushing hazard

Death or serious personal injury

- Make sure that the lifting bracket or strap is tightened before attempting to lift the product. Tighten if necessary.
- Carelessness during the lifting or transportation may cause injury to persons or damage the product.
- Do not stand under or close to the product during and after lifting. Comply with local regulations.

Foundation



CAUTION

Collapsing hazard

Minor or moderate personal injury

- Place support for the crane at a suitable distance from the pit to avoid collapse of the pit hole. Work according to local regulations.

Pump(s) and pump controller(s)



WARNING

Crushing hazard

Death or serious personal injury

- The lifting equipment used for lifting the pump must be rated for the weight of the pump, approved and maintained according to local regulations.

Electrical connection



WARNING

Electric shock

Death or serious personal injury

- The pump or pump controller must be connected to an external emergency stop.



DANGER

Electric shock

Death or serious personal injury

- Switch off the power supply before you start any work on the product.
- Make sure that the power supply cannot be switched on accidentally.

Accessing the products



DANGER

Explosive environment

Death or serious personal injury

- Make sure that explosive gases are evacuated from the pit before working with electrical tools.



DANGER

Suffocation hazard

Death or serious personal injury

- Before entering the pit, make sure it is ventilated according to local regulations. If not, do not enter the pit.



DANGER

Falling hazard

Death or serious personal injury

- When entering the pit, wear a safety harness and use approved equipment for lifting persons up from the pit. Observe local regulations.



WARNING

Crushing of hands

Death or serious personal injury

- Before entering the product, make sure that the cover and safety grid are locked in open position.



CAUTION

Biological hazard

Minor or moderate personal injury

- Use gloves and other suitable personal protection equipment in accordance with local regulations.

Lifting a hookup or free-standing pump



DANGER

Electric shock

Death or serious personal injury

- Switch off the power supply before you start any work on the product. Make sure that the power supply cannot be switched on accidentally.

Lifting the pump on auto coupling



DANGER

Electric shock

Death or serious personal injury

- Switch off the power supply before you start any work on the product.

Repairing a non-return valve and the pipe system



CAUTION

Pressurised system

Minor or moderate personal injury

- Relieve the pressure by draining the system before dismantling the pressure system.



DANGER

Electric shock

Death or serious personal injury

- Remove the fuses or switch off the main switch before dismantling the pressure system. Make sure that the power supply cannot be switched on accidentally.



CAUTION

Biological hazard

Minor or moderate personal injury

- Rinse the pressure system thoroughly with clean water before carrying out maintenance and service.

3. Receiving the product

3.1 Transporting the product



When transporting and handling the product at low temperatures, take into consideration that the product impact resistance is reduced.

- Transport the product in horizontal position and anchor it to the lorry.
- Secure accessories, if any, inside the product.
- Observe the following points during transport and handling:
 - Fix the product securely during transport.
 - Do not dump the product from the lorry.
 - Handle and lift the product according to the local regulations.
 - Use approved straps of textile or similar materials when lifting the product on to or down from a lorry or moving it on the construction site.
 - Point loads must not occur.
 - Do not drag the product over the ground.
 - Do not expose the product to sharp edges.
 - When the product is placed on the ground, make sure the ground is even.



DANGER

Crushing hazard

Death or serious personal injury

- Place the pit on even ground to prevent the pit from overturning.



If the pit is equipped with a lifting point, use it during handling.



Before raising the product into vertical position, remove transport supports and accessories, if any, from the inside of the product.

The customer must provide appropriate lifting equipment at the place of unloading. The weight of the pit is stated on the nameplate.

3.2 Inspecting the product

After transport and before installation, the product must be inspected by the customer.

To complete the inspection, follow the steps below:

1. Check the product for transport damage. Contact the transporter immediately in case of damage.
2. Check that the delivered products correspond to the order.
3. Check the positions and sizes of fittings.
4. Retighten various connections, as they may have become loose during transport.
5. Check that all valves, except the draining valve in the valve chamber, are open.
6. Check other equipment, such as vent pipes.

3.3 Precautions during lifting



DANGER

Crushing hazard

Death or serious personal injury

- Make sure that the lifting bracket or strap is tightened before attempting to lift the product. Tighten if necessary.
- Carelessness during the lifting or transportation may cause injury to persons or damage the product.
- Do not stand under or close to the product during and after lifting. Comply with local regulations.



If the product is equipped with a lifting point, you must use it during handling and lifting.

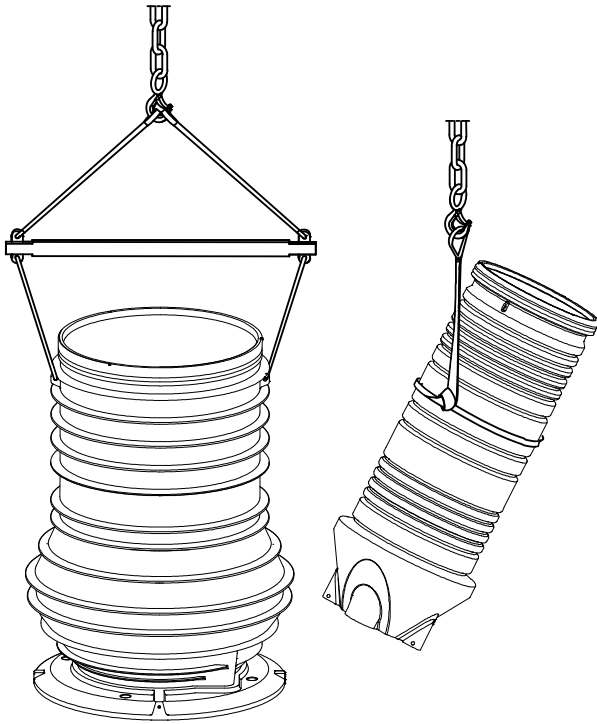


Fig. Lifting the pit

4. Installing the product

4.1 Foundation

CAUTION Collapsing hazard



Minor or moderate personal injury
- Place support for the crane at a suitable distance from the pit to avoid collapse of the pit hole. Work according to local regulations.



Installation of pits must be carried out by an authorised person in accordance with local regulations. Work in or near wastewater pits must be carried out according to local regulations.



The pumping station must be located so that it does not damage other equipment and vice versa.

With reference to DS/EN 1997-1: Eurocode 7: Geotechnical Category 2: We recommend that you complete a geotechnical investigation of the site conditions before you install the pit.

With reference to DS/EN 1997-1: Eurocode 7: Geotechnical Category 1: The geotechnical investigation can be judged to be unnecessary, if the conditions below are fulfilled:

- There is a negligible risk of overall instability, ground movements or adverse ground conditions.
- Comparable local experience shows that the ground conditions are sufficiently uncomplicated.
- There is no excavation below the water table.
- Comparable local experience indicates that the proposed excavation below the water table will be uncomplicated.

In case of doubt, consult a geotechnical specialist.

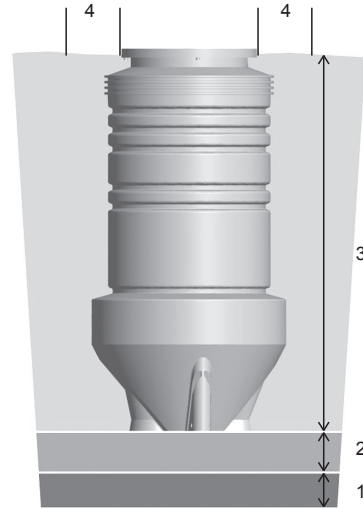


Fig. Schematic installation drawing

Pos.	Description
1	Foundation layer
2	Bedding layer
3	Backfill, compacted in layers of maximum 30 cm
4	Distance of 50 cm from cover where heavy loads must not occur

4.1.1 Foundation layer

- If soil analyses and information about the pit load indicate that the soil is not capable of bearing the weight of the pit, a foundation layer must be prepared.
- The bedding layer on which the pit is placed is not regarded as a foundation layer.
- The foundation layer can be made after the excavation by laying a stable layer of suitable gravel or similar material and compacting it in layers of maximum 30 cm, corresponding to approximately 20 cm after compaction. Such a foundation layer is also required if the excavation has become too deep by mistake.

4.1.2 Bedding layer

- The material for the bedding layer must be compactable and composed so that neither its properties nor the compaction will cause damage to the pit.
- Grains larger than 16 mm must not be present.
- The content of grains between 8 and 16 mm must not exceed 10 %.
- The material must not be frozen.
- Sharp flint etc. must not be present.
- The bedding layer must have a thickness of 10 cm.

4.2 Installing the pit and valve chamber

Before installing the pit or valve chamber, ensure that the foundation and bedding layer have been carried out according to the guidelines in [4.1 Foundation](#).



The valve chamber is only to be installed for PS.R.17 pits.



If the pit is to be installed at a temperature below 0°C, loosen all bolts of the pipe flanges and retighten them when the pit has been installed. This will prevent stress in the pipes.

The following instructions guide you through the installation of the pit and the valve chamber.

- Place the pit or the valve chamber on the bedding layer.
 - Check that the pit or the valve chamber is vertical.
- Carry out the backfilling.

Make sure that the backfilling is carried out according to the guidelines in [4.2.1 Backfill](#).

 - Groundwater must not rise above the bedding layer until the backfilling has been completed.
 - Compact the backfill around the pit up to the lower part of a pipe connection before connecting the pipe.
- Drill holes for the pipes.

Holes for the inlet pipe, electrical connection and venting are to be drilled on site. Before connecting the pipes, check the following:

 - The pipes and gaskets must be clean.
 - The inlet pipe must be properly aligned with the pipe connection.



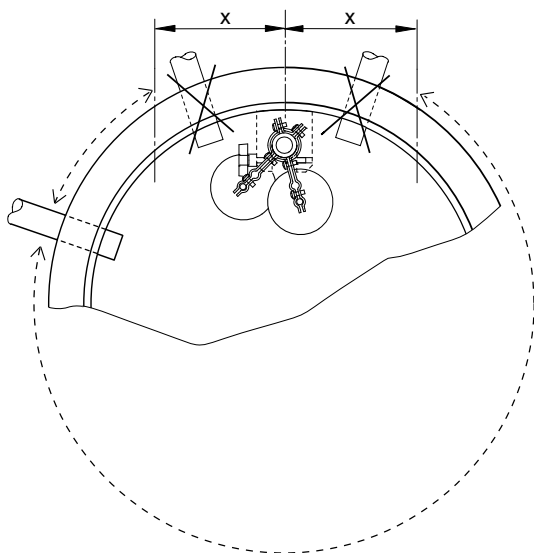
We recommend that you provide the pit with a vent pipe (accessory).



Pipe connections to the pumping station must be laid out according to local regulations.

a. Mark the pit where the inlet is to be.

The pit inlet must not be located in the area of the flow switches, as it will disturb the function of the flow switches. See the figure below.



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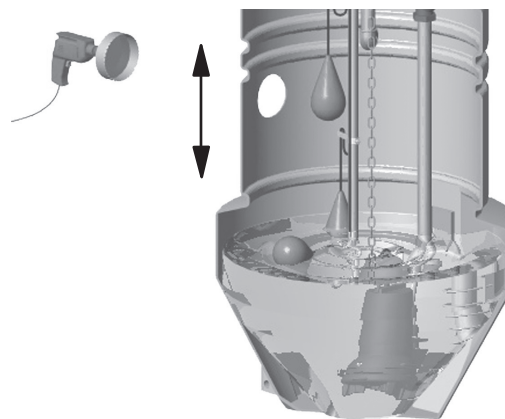
Fig. Location of inlet, $x = 200$ mm

- b. Drill the marked spot with a hole saw. Remove all burrs from the hole.

Sleeve	Pipe diameter	Hole saw [mm]	Borehole tolerance [mm]
LM50/40	40	51	-2/+1
LM50/50	50	60	-2/+1
LM50/63	63	75	-2/+1
LM50/75	75	86	-2/+1
LM50/90	90	102	-2/+1
L965/110	110	127	-0/+2
L965/160	160	177	-0/+2
LM100/200	200	212	-2/+1

PS.R.05 - PS.R.10

We recommend that you drill the inlet hole in the cylindrical, smooth part.



TM069802-1

Fig. Drilling the inlet hole, Ø500-1200

PS.R.17



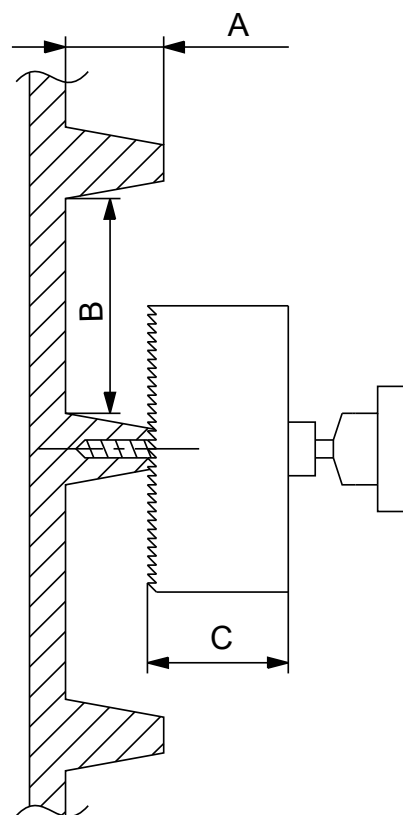
The hole saw must have a cup depth of minimum 90 mm.



The inlet hole must not be drilled in the pit close to pipes, float switches or other parts of the pit installation.

Drilling from the outside

If the hole diameter is bigger than 150 mm or the hole is to be drilled into a rib, drill in the centre of a rib.



TM049441

Fig. Drilling from the outside

Position	Dimension [mm]
A	65
B	150

Position	Dimension [mm]
C	90

Drilling from the inside

Due to the circular pit shape, the pilot drill must extend beyond the cutting edge of the hole saw by at least 23 mm. The position of the hole is not an issue.

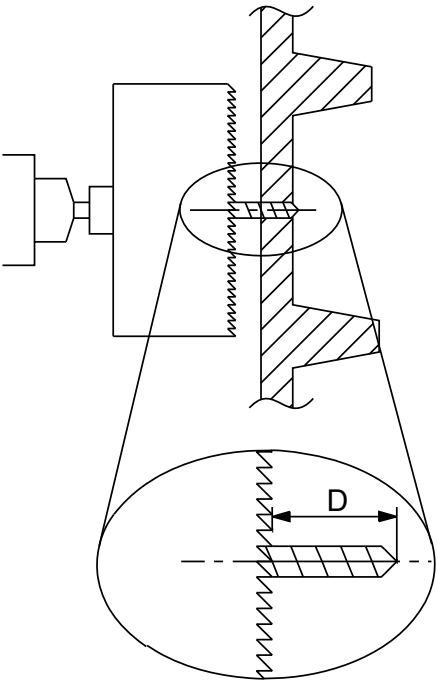


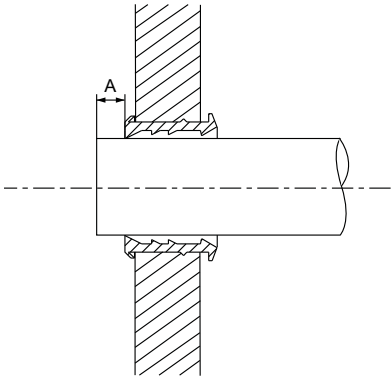
Fig. Drilling from the inside

Position	Dimension [mm]
D	23

- Fit the sleeve.
 - Fold the sleeve and insert it into the hole (from the outside of the pit).
 - Unfold the sleeve so that it lies evenly in the hole.
 - Pull the sleeve outwards until its collar touches the inside of the pit.
 - Lubricate the inside of the sleeve with a water-based lubricant.
- Chamfer the pipe and press it home.

LM50/LM100 sleeves

Push the inlet pipe through the sleeve.



As it must be possible to pull the pump up through the pit, the inlet pipe must not project beyond the sleeve by more than 50 mm (A).

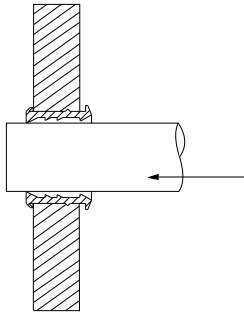


Fig. Pipe with LM50/LM100 sleeve, seen from the side (pit side is to the left)

L965 sleeves

Push the inlet pipe into the sleeve and against the stop.

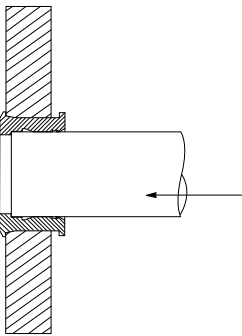
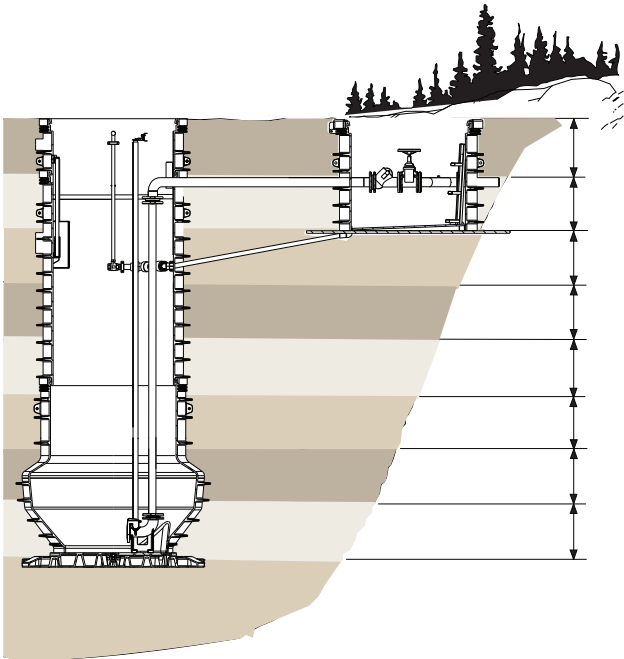


Fig. Pipe with L965 sleeve, seen from the side (pit side is to the left)

Finished installation of pit and valve chamber



Related information

- 4.1 Foundation
 - 4.1.1 Foundation layer
 - 4.1.2 Bedding layer
- 4.2.1 Backfill
- 4.2.2 Uplift prevention

4.2.1 Backfill

When carrying out the backfilling, the following conditions must be fulfilled:

- The backfill must meet the same requirements as those for the bedding layer.
- The backfill must provide sufficient support of the pit on all sides and ensure that the load can be transferred without detrimental point impacts or similar impacts.
- The backfilling must be carried out so that the pit is not damaged or deformed.



During backfilling and compaction, the cover must be fitted on the pit to ensure that the pit is not deformed (becomes oval).

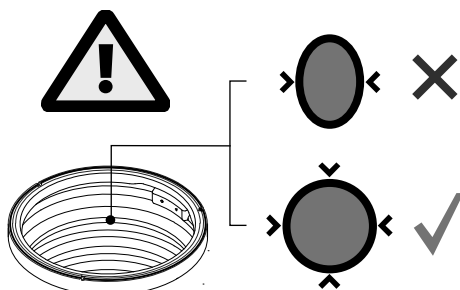


Fig. Even pressure on all sides of the pit

- The backfill must be compacted in layers of maximum 30 cm, corresponding to approximately 20 cm after compaction.



Compact the backfill under the inlet and outlet pipes properly so that they are not exposed to downward loads when the backfill settles.

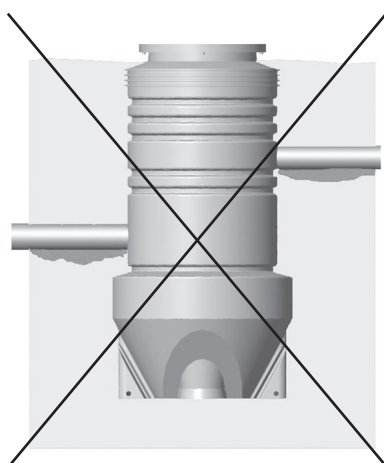


Fig. Insufficient compaction under pipes



If the hole for the pit is drilled, you must pay special attention to the compaction of the backfill to prevent uplift.



Use mechanical compacting equipment and compact the backfill to minimum 97 % proctor density.

- When using compactors near the pit, you must only use either a plate or jumping compactor. The size of the plate compactor must not exceed 25 kN / 150 kg.



Do not use plate compactors at a distance of less than 30 cm from the pit wall.

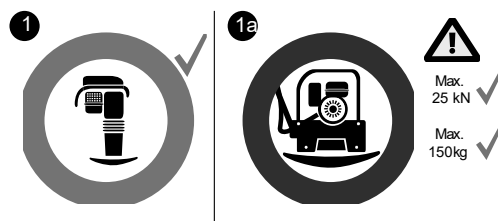


Fig. Allowed compactors

4.2.2 Uplift prevention

The pit design ensures that the pit is protected against uplift if installed correctly (excluded D 800 x 1000 and D 1200 x 1500). Requirements for uplift prevention depend on the geotechnical conditions. They must be determined by the engineer or contractor and are outside Grundfos' area of responsibility.



The requirements below are the minimum requirements. Observe local regulations.

The safety against uplift is determined according to Eurocode EN 1997-1 including national annex, DS/EN 1997-1 DK NA:2015.

The most unfavourable ground water conditions for the pit is when the ground water table is located at terrain level. This situation creates the highest hydrostatic water pressure at the bottom of the pit, and causes the highest reduction in soil weight for the stabilising soil volume.

The sand volume resting on the pit's shoulder is a stabilising force. The sand volume is determined as a cone with a hollow core. The inclination of the cone is illustrated 1:n. The inclination n can be determined based on the friction angle for the design. The characteristic friction angle for sand with a unit weight of 19 kN/m³, compacted to 97 % standard proctor, is estimated to 37 degrees. The design friction angle is 32 degrees which corresponds to an inclination of n = 1.6.

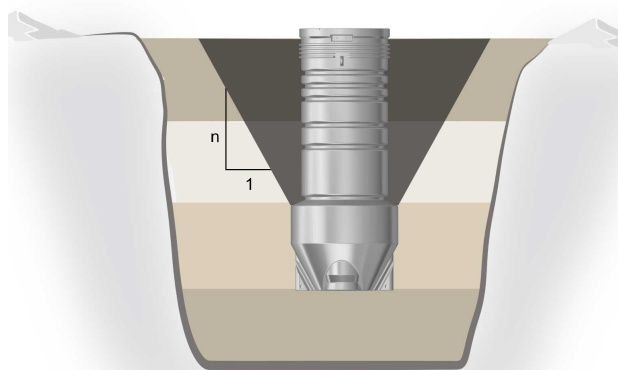


Fig. Stabilising sand volume

4.3 Pump(s) and pump controller(s)

Some products are delivered without the pump(s) installed. For installation and startup of the pump, see the installation and operating instructions of the pump.



Lower the pump(s) carefully into the pit in order to avoid any damage to the pump(s) or the pit.

Fitting the chain

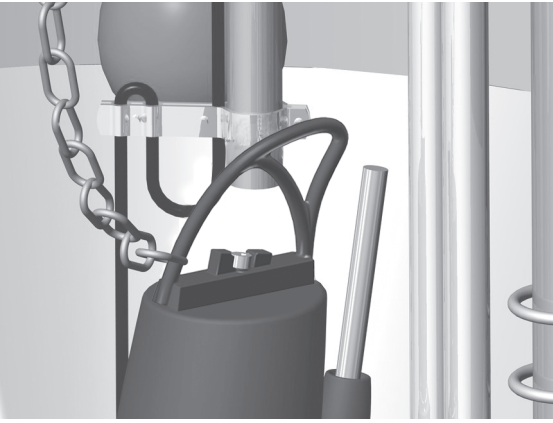
For pumps on auto coupling, we recommend that you fit the chain in the foremost lifting eye of the lifting bracket. The number of lifting eyes depends on the pump.

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TM063603

TM069803

TM072503



TM029341

Fig. Chain fitted to the pump

WARNING

Crushing hazard

Death or serious personal injury

- The lifting equipment used for lifting the pump must be rated for the weight of the pump, approved and maintained according to local regulations.

Chains supplied by Grundfos are marked with maximum load and production date. Do not exceed the maximum load. Maintain the chain(s) according to local regulations. We recommend that you check the chain(s) and shackle(s) supplied by Grundfos at least once a year for cracks, corrosion or other irregularities. If you find any defects, replace the chain(s) and/or shackle(s).

4.4 Level controllers

For installing the level control system, see its installation and operating instructions.



If the pit is equipped with an Auto_{ADAPT}-controlled pump, an external level controller is not needed.

When installing the level controllers, the following points must be observed:

- To prevent air intake and vibrations in the pump, the stop level switch must be fitted in such a way that the pump stops before it sucks in air.
- In order to ensure cooling of pumps without internal stator cooling, the stop level switch must be installed according to the specifications in the installation and operating instructions for the pump.
- In case of one-pump operation, the start level switch must be installed in such a way that the pump starts at the required level. However, the pump must always start before the liquid level reaches the lower edge of the bottom inlet pipe.
- In case of two-pump operation:
 - The start level switches must start pump one earlier.
 - Pump two must start before the liquid reaches the lower edge of the bottom inlet pipe.
- The high-level alarm switch must always be installed about 100 mm above the start level switch. However, the alarm must always be activated before the liquid level reaches the inlet pipe.

For further settings, see the installation and operating instructions of the level controllers.

4.4.1 Start and stop levels

The effective volume of the pit must be large enough so that the number of starts does not exceed the maximum permissible number.

For more information on the maximum number of starts, see the installation and operating instructions for the pump.

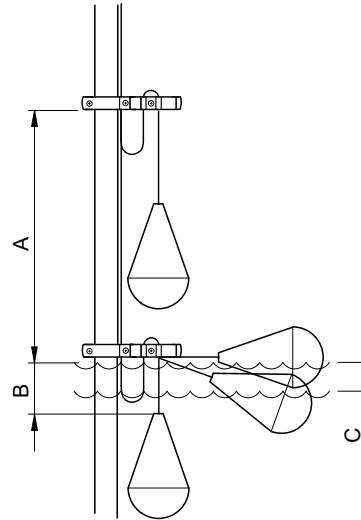
4.4.2 Adjustment of float switches

Float switches

If float switches have been selected, they can be fitted on a tube which can be lifted out of the pit. This ensures easy adjustment of the float switches.



The bottom float switch (stop level) must stop the pump before the liquid level in the pit falls below the minimum level of the pump. For more information, see the installation and operating instructions for the pump.



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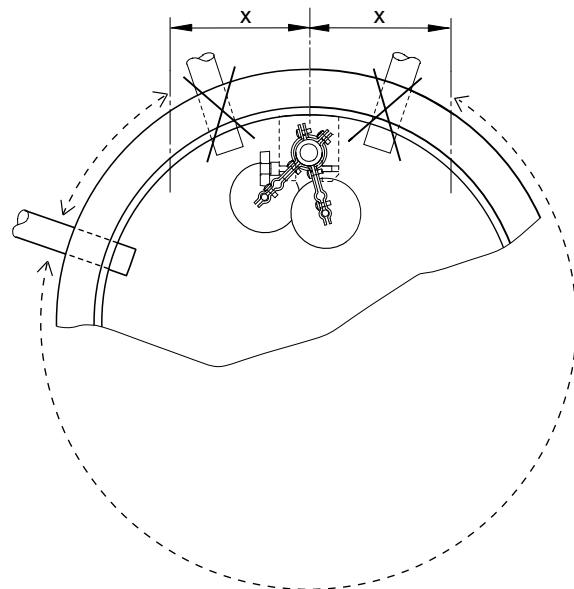
Fig. Adjustment of float switches

A	Minimum 300 mm
B	50 to 100 mm
C	Operating range 110 mm



Distance B must not be too big. Otherwise, the float switch may get stuck in other parts of the installation.

The pit inlet must be located outside of the area shown in the figure below. Otherwise, it will disturb the function of the float switches.



TM028961

Fig. Location of inlet, $x = 200$ mm

4.4.3 Adjustment of electrodes

Electrodes come in a standard length of 1 m. If electrodes are installed for the purpose of level control, their lengths must be adjusted so that they are activated at the correct level.

When you have shortened the electrodes to the correct length, strip 10 to 15 mm of the lower end of the electrodes.

If possible, bend the electrodes to make distance between them. Paper and similar items can get stuck between the electrodes and disturb the signal.

The electrodes come with a 10 m cable.

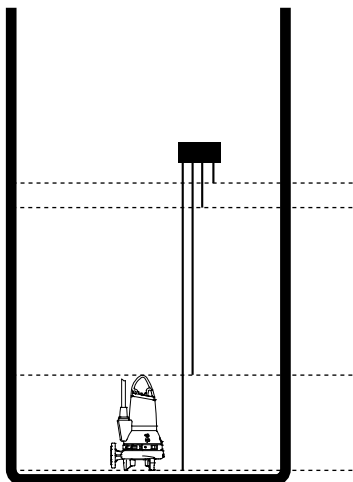


Fig. Electrodes in a pit with one pump

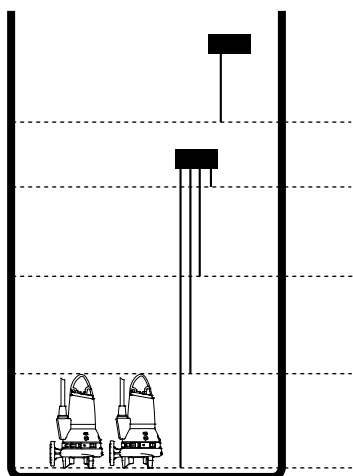


Fig. Electrodes in a pit with two pumps

4.4.4 Air bells

Air bells are pulse generators for the pressure switches of the pump controller and function by means of pneumatic pressure changes. The pressure changes when the liquid level in the air bell changes. The air bells are connected to the pressure switches of the controller by means of pressure hoses.

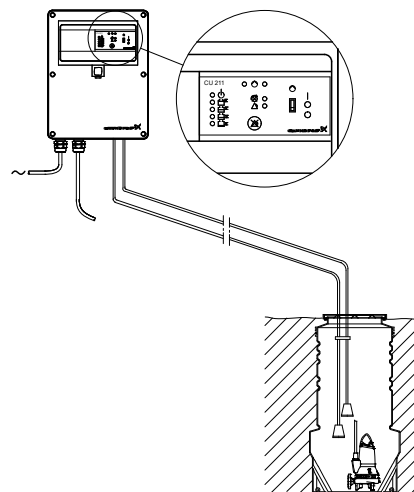


Fig. Air bells in a pit with one pump

4.4.5 Pressure transducer

If a pressure transducer is used for level control, install it in a protective pipe in order to prevent contamination and deposits.

You can use float switches as backup in case the pressure transducer fails to perform. When a float switch is placed in the bottom, it will protect the pump, and when placed in the top, it will prevent overflow.

AutoADAPT pumps do not require pressure transducers since they have built-in pressure sensors.

4.5 Electrical connection

See the installation and operating instructions for the pump and the pump controller.



All electrical connections must be carried out by a qualified electrician in accordance with local regulations.



WARNING

Electric shock

Death or serious personal injury

- The pump or pump controller must be connected to an external emergency stop.



DANGER

Electric shock

Death or serious personal injury

- Switch off the power supply before you start any work on the product.
- Make sure that the power supply cannot be switched on accidentally.



Do not install Grundfos control boxes and the free end of the power cable inside the pumping station.

4.5.1 Cables

The cables for the level switches and the pump(s) can be led into the pit in different ways. Often the cables are led into the pit through a cable entry in the side of the pit. If the pit is equipped with a vent pipe, the vent pipe can be used as a cable entry.



Adjust the length of the power cable by coiling it up on a relief fitting to ensure that the cable is not damaged during operation. Fasten the relief fitting to a suitable hook at the top of the tank. Make sure that the cable is not sharply bent or pinched.



If corrosive gasses are present, we recommend that you use a gas-tight cable entry.

TM028962

TM028962

TM028963

5. Product introduction

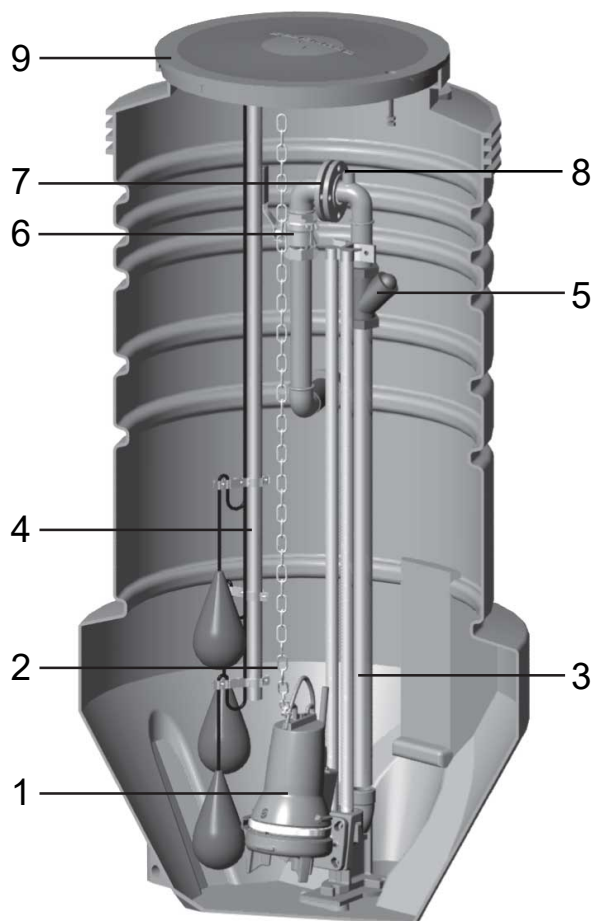
5.1 Product description

Grundfos pumping stations are pre-fabricated pumping stations for collection and pumping of wastewater.

The pit is made of PE-HD (polyethylene) and comes with outlet pipes and valves fitted.

The pipes are made of PE or stainless steel depending on the location of the pit and the pumped liquid, and thus the pump selected.

Wastewater is led into the pit. When the liquid in the pit reaches the maximum liquid level, the pump will start and pump the liquid further on into the sewer system.



TM069807

Fig. Example of a pumping station

Pos.	Description
1	Pump
2	Chain
3	Guide rails
4	Float switch
5	Non-return valve
6	Isolating valve
7	Flange or coupling
8	Connection, ½" internal thread
9	Cover

5.2 Application

Grundfos pumping stations are used for collection and pumping of drainage water, grey wastewater and sewage. For information on each individual pump, controller or sensor, see their respective installation and operating instructions.

5.3 Pumped liquids

Viscosity

The viscosity of the pumped liquid must not exceed the permitted viscosity level of the pump. For further information, see the installation and operating instructions for the pump.

Acids and alkalis

The pump pit is resistant to strong acids and alkalis as well as solvents.

The pump supplied with the pit is normally capable of withstanding pH values between 4 and 10. In case of doubt, please contact Grundfos.

Liquid temperature

The maximum temperature for pumped liquids in the pump pit is 40 °C. For higher temperatures, contact Grundfos.



Select the pump on the basis of the liquid temperature. For further information on each individual pump, see their respective installation and operating instructions.

Density

Maximum 1100 kg/m³.

5.4 Pumping station systems (PS.S)

This section applies only to pumping station systems consisting of Grundfos-approved components mentioned on the nameplate of PS.S.

The pumping station system consists of several elements making the pumping station function as one unit:

- pit
- pump
- pump controller
- level controller
- accessories.

The pumping station system may not always contain all five elements. However, it will always have a pit and a pump. The pump controller and level controller may be incorporated in the pump, while accessories are either added or omitted.

5.5 CE approval of PS.S

PS.S is CE approved in accordance with the following directives and standards:

- EN 2006/42/EC, EU Machinery Directive
- EN/ISO 12100
 - Safety of machinery
 - General principles for design
 - Risk assessment and risk reduction.

In order to ensure safe transport and to fulfil customer requests, the PS.S elements can be assembled on site. However, the CE approval of PS.S is only valid if the following conditions have been fulfilled:

- PS.S has been assembled correctly in accordance with the installation and operating instructions for PS.S and the pit, pump and control systems.
- PS.S contains the Grundfos specified elements stated on the PS.S nameplate. The PS.S nameplate is fitted inside the pit.

If one or more of the components, for example the controllers, in the pumping station system are not supplied by Grundfos, it is not possible to obtain a CE approval.

5.6 Pumping station modular (PS.M)

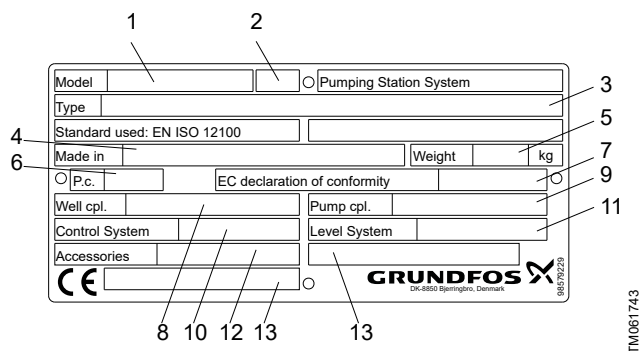
PS.M is used instead of PS.S when the CE approval for the complete system is not needed or some of the components, for example the controllers, in the pumping station are not supplied by Grundfos.

A risk assessment for the complete system must be performed by the customer, if necessary.

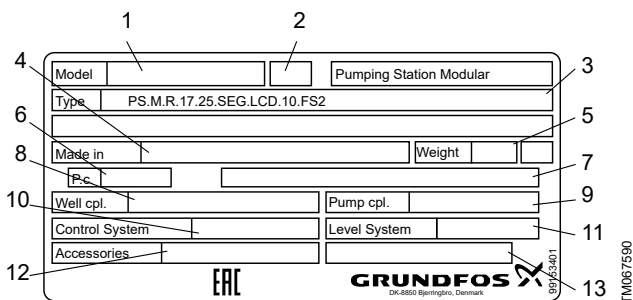
5.7 Identification

5.7.1 Nameplate, PS.S and PS.M

PS.S nameplate



PS.M nameplate



Pos.	Description
1	Product number
2	Production site
3	Type designation
4	Country of origin
5	Weight
6	Production code (year and week)
7	Publication number of the installation and operating instructions
8	Product number, pit
9	Product number, pump
10	Product number, pump controller
11	Product number, level controller
12	Product number(s), accessories
13	Not filled in

5.7.2 Type key, PS.S and PS.M

Example: PS.S.R.17.25.SEG.LCD110.FS2

Code	Explanation	Designation
PS	Grundfos pumping station	System type
S	CE-marked system	
M	Modular system	
R	Rotation-moulded PE	
		Pit type and material

Code	Explanation	Designation
05	500 mm	Pit sump diameter [mm] x 100
08	800 mm	
10	1000 mm	
12	1200 mm	
17	1700 mm	Pit depth [mm] x 100
25	2500 mm	
CC	Unilift CC	
KP	Unilift KP	
AP12	Unilift AP12.50	Pump type
AP35	Unilift AP35, Unilift AP12.40	
AP50	Unilift AP50	
APB	Unilift AP35B, Unilift AP50B	
SEG	SEG	Pump controller
DP/EF	DP (0.6 - 1.5 kW), EF	
DP/SL	DP (2.6 kW), SL1.50.65 and SLV.65.65	
DPK	DPK, DPK.V	
SE/SL	SE/SL	Level controller
X	No Grundfos controller	
BIP	Built into the pump	
CU	Control unit	
LC107	Level controller	Level controller
LC108	Level controller	
LC110	Level controller	
LC115	Level controller	
LCD107	Level controller - two pumps	Pump controller
LCD108	Level controller - two pumps	
LCD110	Level controller - two pumps	
LCD115	Level controller - two pumps	
DC318	Dedicated Controls	Pump controller
DC319	Dedicated Controls	
DCD318	Dedicated Controls - two pumps	
DCD319	Dedicated Controls - two pumps	
BIP	Built into the pump	Level controller
AB2	Two air bells	
AB3	Three air bells	
FS2	Two float switches	
FS3	Three float switches	Level controller
FS4	Four float switches	
EL3	Three electrodes	
EL4	Four electrodes	
EL5	Five electrodes	Level controller
PT	Pressure transducer	
PF1	Pressure transducer and one float switch	
PF2	Pressure transducer and two float switches	

5.7.3 Nameplate, PS.R.05 - PS.R.17

Model	96835168	P10	○
Type	PS.R.17.45.D.DC.SS65.A65.DP/SL		
Standards used	EN 12050-1 or EN 12050-2		
Made in	Denmark	Weight	636 kg
○ P.c.	1325	EU declaration of performance	96235218
  			
Notified body: 0197			

TM063908

Pos. Description

1	Product number
2	Production site
3	Type designation
4	Country of origin
5	Weight
6	Production code (year and week)
7	Publication number of the installation and operating instructions
8	Not filled in

5.7.4 Type key, PS.R

Example: PS.R.17.25.D.GC.SS50.A50.SEG

Code	Explanation	Designation
PS	Pumping station	
R	Rotation-moulded pit	
05	500	
08	800	
10	1000	Pit sump diameter [mm] x 100
12	1200	
17	1700	
25	2500	Pit depth [mm] x 100
S	One pump	Number of pumps
D	Two pumps	
DC	Direct outlet, common	
GC	Goose neck, common	Pipe design
VC	Valve chamber	
SS40	DN 40 (1 1/2")	
SS50	DN 50 (2")	Pipe material and pipe diameter
SS65	DN 65 (2 1/2")	
SS80	DN 80 (3")	Stainless steel
SS100	DN 100 (4")	
PE40	D40 mm (1 1/4")	
PE50	D50 mm (1 1/2")	
PE63	D63 mm (2")	Pipe material and pipe diameter
PE75	D75 mm (2 1/2")	Polyethylene
PE90	D90 mm (3")	
PE110	D110 mm (4")	

Code	Explanation	Designation
A40	DN 40 pump connection	
A50	DN 50 pump connection	
A65	DN 65 pump connection	Installation type
A80	DN 80 pump connection	Auto coupling
A100	DN 100 pump connection	
S	Free-standing pump	Installation type Free-standing pump
H40	DN 40 pump connection	Installation type Hookup auto coupling
KP	Unilift KP, Unilift CC	
AP35	Unilift AP12.40, Unilift AP35	
AP50	Unilift AP12.50, Unilift AP50	
APB	Unilift AP35B, Unilift AP50B	Pump type
SEG	SEG	
DP/EF	DP (0.6 - 1.5 kW) / EF	
DP/SL	DP (2.6 kW) / SL1.50.65 / SLV.65.65	
DPK	DPK, DPK.V	
SE/SL	SE/SL	

1 Maximum 3 m pit depth

6. Service

For servicing the pump or pump controller, see the installation and operating instructions for the product.

It is possible to make a service contract with a local Grundfos company.



We recommend that you make all maintenance and service work when the pump is placed outside the pit.

6.1 Accessing the products



Do not enter and work inside PS.R.05 - PS.R.12.



The pit must be locked to prevent unwanted access.

**DANGER****Explosive environment**

Death or serious personal injury

- Make sure that explosive gases are evacuated from the pit before working with electrical tools.

**DANGER****Suffocation hazard**

Death or serious personal injury

- Before entering the pit, make sure it is ventilated according to local regulations. If not, do not enter the pit.

**DANGER****Falling hazard**

Death or serious personal injury

- When entering the pit, wear a safety harness and use approved equipment for lifting persons up from the pit. Observe local regulations.

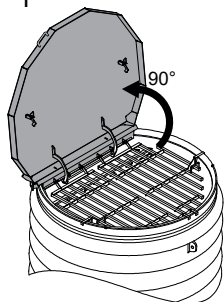


WARNING
Crushing of hands

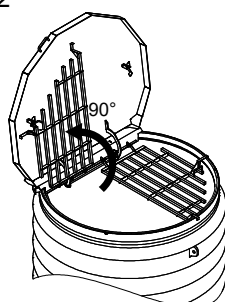
Death or serious personal injury

- Before entering the product, make sure that the cover and safety grid are locked in open position.

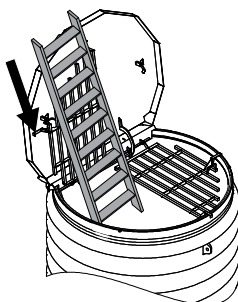
1



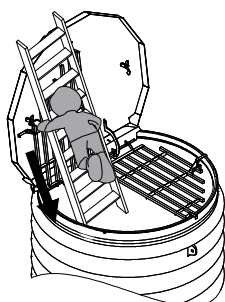
2



3



4



TM060535



Check the ladder supplied by Grundfos, the service platform and the fastening of these at least once per year for cracks, corrosion or other irregularities. Observe local regulations.



All work in pits must be carried out according to local regulations and supervised by at least one person outside the pumping station.



When work is going on in or in the proximity of an open pumping station or valve chamber, place proper warnings signs and correct safety barriers around the pit in order to avoid that persons fall into the pit. The warning signs must be visible from all directions.



If the top opening is up to Ø1000, normal precaution will be sufficient. Openings larger than Ø1000 must be equipped with safety barriers or other safety measures.



If natural light is insufficient, the maintenance persons must use lamps.



CAUTION
Biological hazard

Minor or moderate personal injury

- Use gloves and other suitable personal protection equipment in accordance with local regulations.

6.2 Lifting a hookup or free-standing pump

Make sure that there is enough space above the pit if the outlet pipe is still connected to the pump when lifting the equipment.



Pumps can be lifted by means of a crane, using the lifting points. Use proper slings or chains approved for lifting.



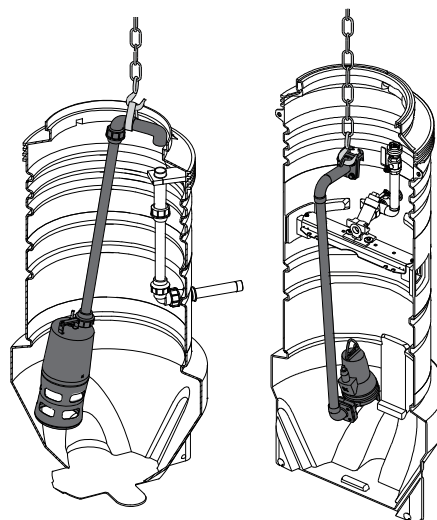
DANGER
Electric shock

Death or serious personal injury

- Switch off the power supply before you start any work on the product. Make sure that the power supply cannot be switched on accidentally.

1. Disconnect the power supply.
2. Disconnect the outlet pipe in the upper bend.
3. Lift the pump and the connected outlet pipe by using the lifting chain. Make sure to guide the outlet pipe.

Example:



TM0702251

6.3 Lifting the pump on auto coupling

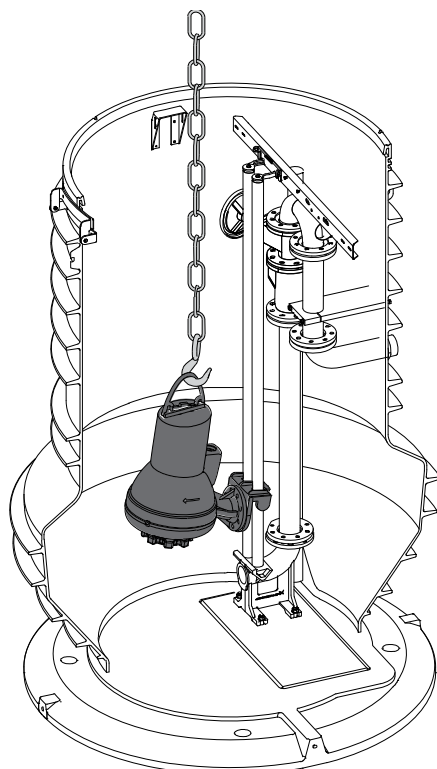
Before lifting the pump, make sure that the outlet valves are closed.



DANGER
Electric shock

Death or serious personal injury

- Switch off the power supply before you start any work on the product.



TM060546

1. Disconnect the power supply.
2. Lift the pump by using the lifting chain.

6.4 Repairing a non-return valve and the pipe system

CAUTION

Pressurised system

Minor or moderate personal injury

- Relieve the pressure by draining the system before dismantling the pressure system.

DANGER

Electric shock

Death or serious personal injury

- Remove the fuses or switch off the main switch before dismantling the pressure system. Make sure that the power supply cannot be switched on accidentally.

CAUTION

Biological hazard

Minor or moderate personal injury

- Rinse the pressure system thoroughly with clean water before carrying out maintenance and service.



Make sure the isolating valves cannot be accidentally opened.

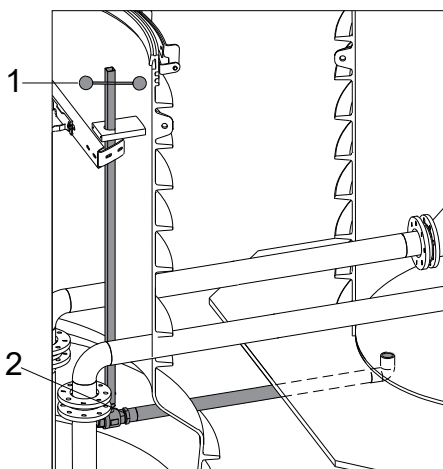


Use a ladder to enter the valve chamber. Always operate valves from inside the valve chamber.



When entering the valve chamber, do not step on pipes or valves.

1. Open the drain valve for the valve chamber by means of the valve handle in the pit in order to empty the sump of the valve chamber.

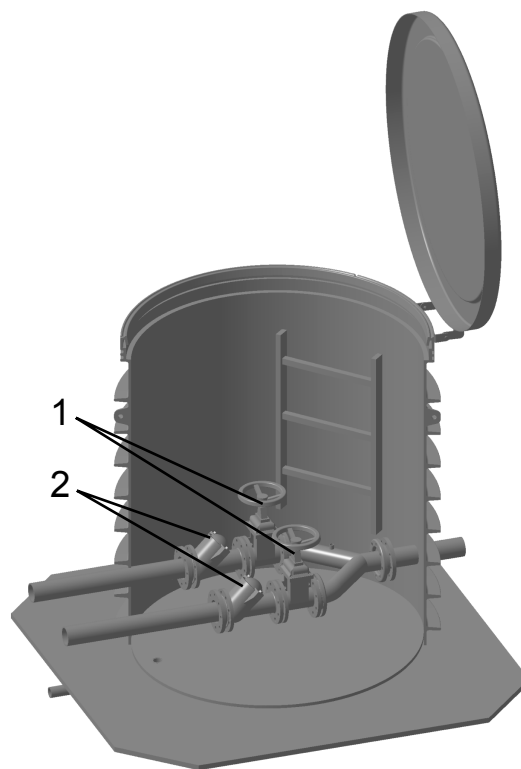


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Fig. Drain valve

1	Valve handle
2	Drain valve

2. Close the isolating valves.

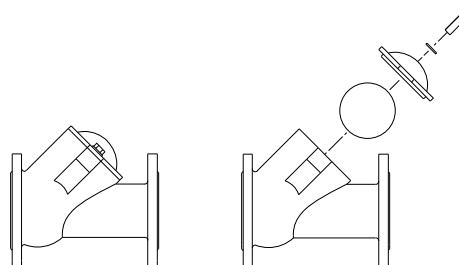


TM069805

Fig. Valve chamber

1	Isolating valves
2	Non-return valves

3. Loosen and remove the two screws on the non-return valve cover.



TM061756

Fig. Non-return valve

4. Replace worn-out valve balls and clean the valves inside.
5. Close the valve cover and tighten the screws.
6. Open the isolating valves.
7. Close the drain valve inside the pit.

7. Disposing of the product

This product or parts of it must be disposed of in an environmentally sound way.

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.
3. If it is not possible to dispose of the product as a complete system, remove the pump and installation equipment and fill and cover the pit. We recommend that you use sand as filling material.



The crossed-out wheellie bin symbol on a product means that it must be disposed of separately from household waste. When a product marked with this symbol reaches its end of life, take it to a collection point designated by the local waste disposal authorities. The separate collection and recycling of such products will help protect the environment and human health.

See also end-of-life information at www.grundfos.com/product-recycling..

8. EC Declaration of conformity PS.S

EC declaration of conformity for Grundfos pumping station system, type PS.S (PS.R, PS.W)

This EC declaration of conformity applies only to Grundfos pumping station system, type PS.S, marked with the CE mark on the nameplate. It is a condition for the validity of the declaration that the pumping station system at any time consists of the correct and Grundfos-approved components and products mentioned in the nameplate of the pumping station system.

We, Grundfos, declare under our sole responsibility that the pumping station system, type PS.S, to which this declaration relates, is in conformity with the Council Directives on the approximation of the laws of the EC member states listed below:

The pumping station system, type PS.S, always consists of a pit, minimum one pump and, depending on the pump type, also a pump controller and a level controller, and possibly accessories, all selected from the list below:

Please find the EC declaration of conformity and/or EU declaration of performance of components and products in the installation and operating instructions delivered with these components and products.

Directive:

Machinery Directive (2006/42/EC).

Standard used: EN ISO 12100:2010.

RoHS Directive: 2011/65/EU and 2015/863/EU.

Standard used: EN 50581:2012.

Components and products:

Pit: PS.R.05, PS.R.08, PS.R.10, PS.R.12, PS.R.17, PS.W.04, PS.W.06, PS.W.08, PS.W.10, PS.W.20

Pump: CC, KP, AP12, AP35, AP50, APB, SEG, DP/EF, DP/SL, DPK, DPK.V, SE/SL, CC-A, KP-A, AP12-A, AP35-A, AP50-A, APB-A, SEG AUTOADAPT, DP/EF AUTOADAPT, DP/SL AUTOADAPT, SE/SL AUTOADAPT

Pump controller: CU, DC/DCD, LC/LCD

Level controller: Air bells, float switches, electrodes, pressure transducer.

Accessories: Ventilation package, vacuum breaker, pressure gauge, sleeves, chains.

This EU declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 96235218).

Székesfehérvár, 14 May 2019

ROBERT-KIS-GMHRKI

Róbert Kis

Senior Manager

Grundfos Holding A/S

Poul Due Jensens Vej 7

8850 Bjerringbro, Denmark

Person authorised to compile the technical file and empowered to sign the EU declaration of conformity.

Declaration of performance

GB: Declaration of performance

EC declaration of performance in accordance with Annex III of Regulation (EU) No 305/2011

(Construction Product Regulation)

1. Unique identification code of the product type:
 - EN 12050-1 or EN 12050-2.
2. Type, batch or serial number or any other element allowing identification of the construction product as required pursuant to Article 11(4):
 - PUST/PS.R pumping stations marked with EN 12050-1 or EN 12050-2 on the nameplate.
3. Intended use or uses of the construction product, in accordance with the applicable harmonised technical specification, as foreseen by the manufacturer:
 - Pumping stations for pumping of wastewater containing faecal matter marked with EN 12050-1 on the nameplate.
 - Pumping stations for pumping of faecal-free wastewater marked with EN 12050-2 on the nameplate.
4. Name, registered trade name or registered trade mark and contact address of the manufacturer as required pursuant to Article 11(5):
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Denmark.
5. NOT RELEVANT.
6. System or systems of assessment and verification of constancy of performance of the construction product as set out in Annex V:
 - System 3.
7. In case of the declaration of performance concerning a construction product covered by a harmonised standard:
 - TÜV Rheinland LGA Products GmbH, identification number: 0197.
Performed test according to EN 12050-1 or EN 12050-2 under system 3. (description of the third party tasks as set out in Annex V)
 - Certificate number: LGA-Certificate No 7310150. Type-tested and monitored. LGA test report No. 7313401-01a.
8. NOT RELEVANT.
9. Declared performance: The products covered by this declaration of performance are in compliance with the essential characteristics and the performance requirements as described in the following:
 - Standards used: EN 12050-1:2001 or EN 12050-2:2000.
10. The performance of the product identified in points 1 and 2 is in conformity with the declared performance in point 9.

CZ: Prohlášení o vlastnostech

Prohlášení o vlastnostech ES v souladu s Dodatkem III předpisu (EU) č. 305/2011

(Předpis pro stavební výrobky)

1. Jedinečný identifikační kód typu výrobku:
 - EN 12050-1 nebo EN 12050-2.
2. Typ, dávka nebo výrobní číslo nebo jakýkoliv prvek umožňující identifikaci stavebního výrobku podle požadavku Článku 11(4):
 - PUST/PS.R Čerpací stanice odpadních vod s označením EN 12050-1 nebo EN 12050-2 na typovém štítku.
3. Zamýšlená použití stavebního výrobku v souladu s příslušnou harmonizovanou technickou specifikací výrobce:
 - Čerpací stanice odpadních vod s fekáliemi s označením EN 12050-1 na typovém štítku.
 - Čerpací stanice odpadních vod bez fekálií s označením EN 12050-2 na typovém štítku.
4. Název, registrovaný obchodní název nebo registrovaná ochranná známka a kontaktní adresa výrobce podle požadavku Článku 11(5):
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Dánsko.
5. NESOUVISÍ.
6. Systém nebo systémy posuzování a ověřování stálosti vlastností stavebního výrobku podle ustanovení Dodatku V:
 - Systém 3.
7. V případě prohlášení o vlastnostech stavebního výrobku zahrnutého v harmonizované normě:
 - TÜV Rheinland LGA Products GmbH, identifikační číslo: 0197.
Proveden test podle EN 12050-1 nebo EN 12050-2 v systému 3. (popis úkolů třetí strany podle ustanovení Dodatku V)
 - Číslo certifikátu: Certifikát LGA č. 7310150. Typ testován a monitorován. Zkušební protokol LGA č. 7313401-01a.
8. NESOUVISÍ.
9. Prohlašované vlastnosti: Výrobky uvedené v tomto Prohlášení o vlastnostech jsou v souladu se základními charakteristikami a požadavky na vlastnosti, jak je popsáno níže:
 - Použité normy: EN 12050-1:2001 nebo EN 12050-2:2000.
10. Vlastnosti výrobku uvedeného v bodech 1 a 2 v souladu s prohlašovanými vlastnostmi v bodě 9.

DK: Ydeevnedeklaration

EU-ydeevnedeklaration i henhold til bilag III af forordning (EU) nr. 305/2011 (Byggevarerforordningen)

1. Varetypens unikke identifikationskode:
 - EN12050-1 eller EN 12050-2.
2. Type-, parti- eller serienummer eller en anden form for angivelse ved hjælp af hvilken byggevaren kan identificeres som krævet i henhold til artikel 11, stk. 4:
 - PUST/PS.R pumpestationer der er mærket med EN 12050-1 eller EN 12050-2 på typeskiltet.
3. Byggevarens tilsigtede anvendelse eller anvendelser i overensstemmelse med den gældende harmoniserede tekniske specifikation som påtænkt af fabrikanten:
 - Pumpestationer til pumpning af spildevand med fækalier der er mærket med EN 12050-1 på typeskiltet.
 - Pumpestationer til pumpning af fækaliefrit spildevand der er mærket med EN 12050-2 på typeskiltet.
4. Fabrikantens navn, registrerede firmabetegnelse eller registrerede varemærke og kontaktadresse som krævet i henhold til artikel 11, stk. 5:
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Danmark.
5. IKKE RELEVANT.
6. Systemet eller systemerne til vurdering og kontrol af at byggevarens ydeevne er konstant, jf. bilag V:
 - System 3.
7. Hvis ydeevnedeklarationen vedrører en byggevarer der er omfattet af en harmoniseret standard:
 - TÜV Rheinland LGA Products GmbH, identifikationsnummer: 0197.
Udført test i henhold til EN 12050-1 eller EN 12050-2 efter system 3. (beskrivelse af tredjepartsopgaverne, jf. bilag V).
 - Certifikatnummer: LGA-certifikat nr. 7310150. Typetestet og overvåget. LGA-testrapport nr. 7313401-01a.
8. IKKE RELEVANT.
9. Deklareret ydeevne: De produkter der er omfattet af denne ydeevnedeklaration, er i overensstemmelse med de væsentlige egenskaber og ydelseskrav der er beskrevet i følgende:
 - Anvendte standarder: EN 12050-1:2001 eller EN 12050-2:2000.
10. Ydeevnen for den byggevarer der er anført i punkt 1 og 2, er i overensstemmelse med den deklarerede ydeevne i punkt 9.

DE: Leistungserklärung

EC-Leistungserklärung gemäß Anhang III der Verordnung (EU) Nr. 305/2011 (Bauprodukte-Verordnung)

1. Einmalige Kennnummer des Produkttyps:
 - EN 12050-1 oder EN 12050-2.
2. Typ, Charge, Seriennummer oder jedes andere Element, das eine Identifizierung des Bauprodukts erlaubt, wie in Artikel 11 (4) vorgeschrieben.
 - PUST/PS.R pumpstationen, auf dem Typenschild mit EN 12050-1 oder EN 12050-2 gekennzeichnet.
3. Verwendungszweck oder Verwendungszwecke des Bauprodukts, gemäß den geltenden harmonisierten technischen Spezifikationen, wie vom Hersteller vorgesehen:
 - Pumpstationen für die Förderung von fäkalienhaltigem Abwasser, auf dem Typenschild mit EN 12050-1 gekennzeichnet.
 - Pumpstationen für die Förderung von fäkalienfreiem Abwasser, auf dem Typenschild mit EN 12050-2 gekennzeichnet.
4. Name, eingetragener Markenname oder eingetragenes Warenzeichen und Kontaktanschrift des Herstellers, wie in Artikel 11(5) vorgeschrieben.
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Dänemark
5. NICHT RELEVANT.
6. System oder Systeme zur Bewertung und Überprüfung der Leistungsbeständigkeit des Bauprodukts gemäß Anhang V:
 - System 3.
7. Bei der Leistungserklärung bezüglich eines von einer harmonisierten Norm erfassten Bauprodukts:
 - TÜV Rheinland LGA Products GmbH, Kennnummer: 0197.
Vorgenommene Prüfung gemäß EN 12050-1 oder EN 12050-2 unter Anwendung von System 3. (Beschreibung der Aufgaben von unabhängigen Dritten gemäß Anhang V)
 - Zertifikatnummer: LGA-Zertifikatnr. 7310150. Typgeprüft und überwacht. LGA-Prüfberichtnr. 7313401-01a.
8. NICHT RELEVANT.
9. Erklärte Leistung: Die von dieser Leistungserklärung erfassten Produkte entsprechen den grundlegenden Charakteristika und Leistungsanforderungen, wie im Folgenden beschrieben:
 - Angewendete Normen: EN 12050-1:2001 oder EN 12050-2:2000.
10. Die Leistung des in Punkt 1 und 2 genannten Produkts entspricht der in Punkt 9 erklärten Leistung.

ES: Declaración de prestaciones

Declaración CE de prestaciones conforme al Anexo III del Reglamento (EU) n.º 305/2011

(Reglamento de productos de construcción)

- Código de identificación único del tipo de producto:
 - EN 12050-1 o EN 12050-2.
- Tipo, lote o número de serie, o cualquier otro elemento que facilite la identificación del producto de construcción de acuerdo con los requisitos establecidos en el Artículo 11(4):
 - PUST/PS.R Estaciones de bombeo en cuya placa de características figuren las normas EN 12050-1 o EN 12050-2.
- Uso o usos previstos del producto de construcción, conforme a la especificación técnica armonizada correspondiente, según lo previsto por el fabricante:
 - Estaciones de bombeo para el bombeo de aguas residuales que contengan materia fecal en cuya placa de características figure la norma EN 12050-1.
 - Estaciones de bombeo para el bombeo de aguas residuales que no contengan materia fecal en cuya placa de características figure la norma EN 12050-2.
- Nombre, nombre comercial registrado o marca comercial registrada y domicilio de contacto del fabricante de acuerdo con los requisitos establecidos en el Artículo 11(5):
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Dinamarca.
- NO CORRESPONDE.
- Sistema o sistemas de evaluación y verificación de la continuidad de las prestaciones del producto de construcción, de acuerdo con lo establecido en el Anexo V.
 - Sistema 3.
- Si la declaración de prestaciones concierne a un producto de construcción cubierto por una norma armonizada:
 - TÜV Rheinland LGA Products GmbH, número de identificación: 0197.
Ensayo ejecutado según las normas EN 12050-1 o EN 12050-2, sistema 3. (Descripción de las tareas de las que deben responsabilizarse otras partes de acuerdo con lo establecido en el Anexo V).
 - Número de certificado: Certificado LGA n.º 7310150. Tipo sometido a ensayo y monitorizado. Informe de prueba LGA n.º 7313401-01a.
- NO CORRESPONDE.
- Prestaciones declaradas: Los productos que cubre esta declaración de prestaciones satisfacen las características fundamentales y requisitos en materia de prestaciones descritos en:
 - Normas aplicadas: EN 12050-1:2001 o EN 12050-2:2000.
- Las prestaciones del producto indicado en los puntos 1 y 2 cumplen lo declarado en el punto 9.

FI: Suoritustasoilmoitus

EY-suoritustasoilmoitus laadittu asetuksen 305/2011/EU liitteen III mukaisesti (Rakennustuoteasetus)

- Tuotetyypin yksilöllinen tunnistus:
 - EN 12050-1 tai EN 12050-2.
- Tyyppi-, erä- tai sarjanumero tai muu merkintä, jonka ansiosta rakennustuotteet voidaan tunnistaa, kuten 11 artiklan 4 kohdassa edellytetään:
 - PUST/PS.R Pumppaamot, joiden arvokilvessä on merkintä EN 12050-1 tai EN 12050-2.
- Valmistajan ennakoima, sovellettavan yhdenmukaistetun teknisen eritelmän mukainen rakennustuotteen aiottu käyttötarkoitus tai -tarkoitukset:
 - Pumppaamot ulosteperäistä materiaalia sisältävien jätevesien pumppaukseen. Arvokilvessä on merkintä EN 12050-1.
 - Pumppaamot sellaisten jätevesien pumppaukseen, jotka eivät sisällä ulosteperäistä materiaalia. Arvokilvessä on merkintä EN 12050-2.
- Valmistajan nimi, rekisteröity kaupp nimi tai tavaramerkki sekä osoite, josta valmistajaan saa yhteyden, kuten 11 artiklan 5 kohdassa edellytetään:
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Tanska.
- EI TARVITA.
- Rakennustuotteen suoritusasteen pysyvyyden arviointi- ja varmennusjärjestelmä(t) liitteen V mukaisesti:
 - Järjestelmä 3.
- Kun kyse on yhdenmukaistetun standardin piiriin kuuluvan rakennustuotteen suoritusasteilmoituksesta:
 - TÜV Rheinland LGA Products GmbH, tunnistenumero: 0197.
Testaus suoritettu standardien EN 12050-1 tai EN 12050-2 ja järjestelmän 3 mukaisesti. (Liitteessä V esitettyjä kolmannen osapuolen tehtävien kuvauksia noudattaen.)
 - Sertifikaatin numero: LGA-sertifikaatti nro 7310150. Tyyppitestattu ja valvottu. LGA-testausseloste nro 7313401-01a.
- EI TARVITA.
- Ilmoitetut suoritusasteet: Tähän suoritusasteilmoitukseen kuuluvien tuotteiden perusominaisuudet ja suoritusastevaatimukset:
 - Sovellettavat standardit: EN 12050-1:2001 tai EN 12050-2:2000.
- Kohdissa 1 ja 2 yksilöidyn tuotteen suoritusasteet ovat kohdassa 9 ilmoitettujen suoritusasteojen mukaiset.

FR: Déclaration des performances

Déclaration des performances CE conformément à l'Annexe III du Règlement (UE) n° 305/2011

(Règlement Produits de Construction)

1. Code d'identification unique du type de produit :
 - EN 12050-1 ou EN 12050-2.
2. Numéro de type, de lot ou de série ou tout autre élément permettant l'identification du produit de construction comme l'exige l'Article 11(4) :
 - PUST/PS.R Stations de pompage marquées EN 12050-1 ou EN 12050-2 sur la plaque signalétique.
3. Usage(s) prévu(s) du produit de construction conformément à la spécification technique harmonisée applicable comme indiqué par le fabricant :
 - Stations de pompage pour le pompage des effluents contenant des matières fécales marquées EN 12050-1 sur la plaque signalétique.
 - Stations de pompage pour le pompage des effluents exempts de matières fécales marquées EN 12050-2 sur la plaque signalétique.
4. Nom, nom de commerce déposé ou marque commerciale déposée et adresse du fabricant comme l'exige l'Article 11(5) :
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Danemark.
5. NON APPLICABLE.
6. Système ou systèmes d'attestation et de vérification de la constance des performances du produit de construction comme stipulé dans l'Annexe V :
 - Système 3.
7. En cas de déclaration des performances d'un produit de construction couvert par une norme harmonisée :
 - TÜV Rheinland LGA Products GmbH, numéro d'identification : 0197.
Test effectué conformément aux normes EN 12050-1 ou EN 12050-2 selon le système 3. (description des tâches de tierce partie comme stipulé dans l'Annexe V)
 - Numéro de certificat : Certificat LGA n° 7310150. Contrôlé et homologué. Rapport de test LGA n° 7313401-01a.
8. NON APPLICABLE.
9. Performances déclarées : Les produits couverts par cette déclaration des performances sont conformes aux caractéristiques essentielles et aux exigences de performances décrites par la suite :
 - Normes utilisées : EN 12050-1:2001 ou EN 12050-2:2000.
10. Les performances du produit identifié aux points 1 et 2 sont conformes aux performances déclarées au point 9.

IT: Dichiarazione di prestazioni

Dichiarazione CE di prestazioni in conformità all'all. III del Regolamento (UE) n. 305/2011

(regolamento sui prodotti da costruzione)

1. Codice identificativo esclusivo del tipo di prodotto:
 - EN 12050-1 oppure EN 12050-2.
2. Tipo, lotto o numero di serie o qualsiasi altro elemento che consenta l'identificazione del prodotto da costruzione come necessario secondo l'art. 11(4):
 - PUST/PS.R Stazioni di pompaggio, marcate con EN 12050-1 oppure EN 12050-2 sulla targa dei dati identificativi.
3. Utilizzo o utilizzi previsti del prodotto da costruzione, in accordo alla specifica tecnica armonizzata pertinente, come previsto dal fabbricante:
 - Stazioni di pompaggio di acque reflue contenenti materiali fecali, marcate con EN 12050-1 sulla targa dei dati identificativi.
 - Stazioni di pompaggio di acque reflue non contenenti materiali fecali, marcate con EN 12050-2 sulla targa dei dati identificativi.
4. Denominazione, denominazione commerciale registrata o marchio registrato e indirizzo di contatto del fabbricante secondo l'art. 11(5):
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Danimarca.
5. NON RILEVANTE.
6. Sistema o sistemi di valutazione e verifica della costanza delle prestazioni del prodotto da costruzione come definito sub all. V:
 - Sistema 3.
7. In caso di dichiarazione di prestazioni concernente un prodotto da costruzione conforme a una norma armonizzata:
 - TÜV Rheinland LGA Products GmbH, numero d'identificazione: 0197.
Test eseguito secondo EN 12050-1 oppure EN 12050-2 con il sistema 3. (descrizione delle mansioni di terzi come definito sub all. V)
 - Numero certificato: N. certificato LGA 7310150. Testato per il tipo e monitorato. N. rapporto di prova LGA 7313401-01a.
8. NON RILEVANTE.
9. Prestazioni dichiarate: I prodotti coperti dalla presente dichiarazione di prestazione sono conformi alle caratteristiche essenziali ed ai requisiti di prestazioni descritti dove segue:
 - Norme applicate: EN 12050-1:2001 oppure EN 12050-2:2000.
10. Le prestazioni del prodotto identificato ai punti 1 e 2 sono conformi alle prestazioni dichiarate al punto 9.

NL: Prestatieverklaring

Prestatieverklaring van EG in overeenstemming met Bijlage III van verordening (EU) nr. 305/2011 (Bouwproductenverordening)

1. Unieke identificatiecode van het producttype:
 - EN 12050-1 of EN 12050-2.
2. Type-, batch- of serienummer of enig ander element dat identificatie van het bouwproduct mogelijk maakt zoals vereist conform artikel 11(4):
 - PUST/PS.R Pompstations gemarkeerd met EN 12050-1 of EN 12050-2 op het typeplaatje.
3. Beoogde toepassing of toepassingen van het bouwproduct, in overeenstemming met de van toepassing zijnde geharmoniseerde technische specificatie, zoals voorzien door de fabrikant:
 - Pompstations voor het verpompen van afvalwater dat fecale materie bevat gemarkeerd met EN 12050-1 op het typeplaatje.
 - Pompstations voor het verpompen van afvalwater dat geen fecale materie bevat gemarkeerd met EN 12050-2 op het typeplaatje.
4. Naam, gedeponeerde handelsnaam of gedeponeerd handelsmerk en contactadres van de fabrikant zoals vereist conform artikel 11(5):
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Denemarken.
5. NIET RELEVANT.
6. Systeem of systemen voor beoordeling en verificatie van constantheid van prestaties van het bouwproduct zoals beschreven in Bijlage V:
 - Systeem 3.
7. In het geval van de prestatieverklaring voor een bouwproduct dat onder een geharmoniseerde norm valt:
 - TÜV Rheinland LGA Products GmbH, identificatienummer: 0197.
Uitgevoerde test conform EN 12050-1 of EN 12050-2 onder systeem 3. (beschrijving van de externe taken zoals beschreven in Bijlage V)
 - Certificaatnummer: LGA-certificaatnr. 7310150. Type getest en bewaakt. LGA-testrapportnr. 7313401-01a.
8. NIET RELEVANT.
9. Verklaarde prestatie: De producten die vallen onder deze prestatieverklaring zijn in overeenstemming met de essentiële eigenschappen en de prestatievereisten zoals beschreven in het volgende:
 - Gebruikte normen: EN 12050-1:2001 of EN 12050-2:2000.
10. De prestaties van het product dat is geïdentificeerd in punten 1 en 2 zijn in overeenstemming met de verklaarde prestaties in punt 9.

PL: Deklaracja właściwości użytkowych

według załącznika III do dyrektywy (WE) nr 305/2011 w/s wprowadzania do obrotu wyrobów budowlanych

1. Niepowtarzalny kod identyfikacyjny typu wyrobu:
 - EN 12050-1 lub EN 12050-2.
2. Numer typu, partii lub serii lub jakiegokolwiek inny element umożliwiający identyfikację wyrobu budowlanego, wymagany zgodnie z art. 11 ust. 4:
 - PUST/PS.R Przepompownie oznaczone na tabliczce znamionowej kodem EN 12050-1 lub EN 12050-2.
3. Przewidziane przez producenta zamierzone zastosowanie lub zastosowania wyrobu budowlanego zgodnie z mającą zastosowanie zharmonizowaną specyfikacją techniczną:
 - Przepompownie do pompowania ścieków zawierających fekalia, oznaczone na tabliczce znamionowej kodem EN 12050-1.
 - Przepompownie do pompowania ścieków bez fekalii, oznaczone na tabliczce znamionowej kodem EN 12050-2.
4. Nazwa, zastrzeżona nazwa handlowa lub zastrzeżony znak towarowy oraz adres kontaktowy producenta, wymagany zgodnie z art. 11 ust. 5:
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Dania.
5. NIE DOTYCZY.
6. System lub systemy oceny i weryfikacji stałości właściwości użytkowych wyrobu budowlanego określone w załączniku V:
 - System 3.
7. W przypadku deklaracji właściwości użytkowych dotyczącej wyrobu budowlanego objętego normą zharmonizowaną:
 - Jednostka certyfikująca TÜV Rheinland LGA Products GmbH, numer identyfikacyjny: 0197.
przeprowadziła badanie określone w EN 12050-1 lub EN 12050-2, w systemie 3 i wydała certyfikat. (opis zadań strony trzeciej, określonych w załączniku V)
 - Nr certyfikatu: certyfikat LGA nr 7310150. Certyfikat badania typu i stałości właściwości użytkowych. Raport z testów LGA nr 7313401-01a.
8. NIE DOTYCZY.
9. Deklarowane właściwości użytkowe: Wyroby, których dotyczy niniejsza deklaracja właściwości użytkowych są zgodne z zasadniczymi charakterystykami i wymaganiami określonymi w następujących normach:
 - Zastosowane normy: EN 12050-1:2001 lub EN 12050-2:2000.
10. Właściwości użytkowe wyrobu określone w pkt 1 i 2 są zgodne z właściwościami użytkowymi deklarowanymi w pkt 9.

PT: Declaração de desempenho

Declaração de desempenho CE, em conformidade com o Anexo III do Regulamento (UE) N.º 305/2011

(Regulamento de Produtos da Construção)

1. Código de identificação exclusivo do tipo de produto:
 - EN 12050-1 ou EN 12050-2.
2. Tipo, lote ou número de série ou qualquer outro elemento que permita a identificação do produto de construção, em conformidade com o Artigo 11(4):
 - PUST/PS.R Estações de bombeamento com a indicação EN 12050-1 ou EN 12050-2 na chapa de características.
3. Utilização ou utilizações prevista(s) do produto de construção, em conformidade com a especificação técnica harmonizada aplicável, conforme previsto pelo fabricante:
 - Estações de bombeamento para bombeamento de águas residuais com conteúdo de matéria fecal com a indicação EN 12050-1 na chapa de características.
 - Estações de bombeamento para bombeamento de águas residuais sem matéria fecal com a indicação EN 12050-2 na chapa de características.
4. Nome, nome comercial registado ou marca registada e endereço de contacto do fabricante, em conformidade com o Artigo 11(5):
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Dinamarca.
5. NÃO RELEVANTE.
6. Sistema ou sistemas de avaliação e verificação da regularidade do desempenho do produto de construção, conforme definido no Anexo V:
 - Sistema 3.
7. Em caso de declaração de desempenho referente a um produto de construção abrangido por uma norma harmonizada:
 - TÜV Rheinland LGA Products GmbH, número de identificação: 0197.
Teste realizado em conformidade com EN 12050-1 ou EN 12050-2 ao abrigo do sistema 3. (descrição das tarefas de partes terceiras, conforme definido no Anexo V)
 - Número do certificado: Certificado LGA N.º 7310150. Testado e monitorizado. Relatório de teste LGA N.º 7313401-01a.
8. NÃO RELEVANTE.
9. Desempenho declarado: Os produtos abrangidos por esta declaração de desempenho cumprem as características essenciais e os requisitos de desempenho conforme descritos em:
 - Normas utilizadas: EN 12050-1:2001 ou EN 12050-2:2000.
10. O desempenho do produto identificado nos pontos 1 e 2 encontra-se em conformidade com o desempenho declarado no ponto 9.

RO: Declarație de performanță

Declarație CE de performanță în conformitate cu anexa III a Regulamentului (UE) nr 305/2011

(reglementare privind produsele pentru construcții)

1. Cod unic de identificare a tipului de produs:
 - EN 12050-1 sau EN 12050-2.
2. Tipul, lotul sau seria, sau orice alt element care permite identificarea produsului pentru construcții după cum este necesar în conformitate cu articolul 11 (4):
 - Stații de ridicare PUST/PS.R marcate cu EN 12050-1 sau EN 12050-2 pe placa de identificare.
3. Utilizarea sau utilizările preconizate ale produsului pentru construcții, în conformitate cu specificația tehnică armonizată aplicabilă, astfel cum este prevăzută de către producător:
 - Stații de pompare pentru pomparea apei uzate conținând materii fecale, marcate cu EN 12050-1 pe placa de identificare.
 - Stații de pompare pentru pomparea apei uzate fără materii fecale, marcate cu EN 12050-2 pe placa de identificare.
4. Numele, denumirea comercială înregistrată sau marca înregistrată și adresa de contact a fabricantului cerute conform cu articolului 11 (5):
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Danemarca.
5. NU ESTE RELEVANT.
6. Sistemul sau sistemele de evaluare și verificare a constanței performanței produsului pentru construcții astfel cum este prevăzută în anexa V:
 - Sistemul 3.
7. În cazul declarației de performanță pentru un produs pentru construcții specificat într-un standard armonizat:
 - TÜV Rheinland LGA Products GmbH, număr de identificare: 0197.
Test efectuat conform EN 12050-1 sau EN 12050-2 potrivit sistemului 3. (descrierea sarcinilor terței părți așa cum este prevăzută în anexa V)
 - Numărul certificatului: LGA-Certificat nr. 7310150. Tip testat și monitorizat. LGA raport de testare nr. 7313401-01a.
8. NU ESTE RELEVANT.
9. Performanță declarată: Produsele specificate de această declarație de performanță sunt în conformitate cu caracteristicile esențiale și cerințele de performanță descrise în cele ce urmează:
 - Standarde utilizate: EN 12050-1:2001 sau EN 12050-2:2000.
10. Performanța produsului identificat la punctele 1 și 2 este în conformitate cu performanța declarată la punctul 9.

SK: Vyhlásenie o parametroch

Vyhlásenie o parametroch v súlade s prílohou III nariadenia (EÚ) č. 305/2011 (Nariadenie o stavebných výrobkoch)

- Jedinečný identifikačný kód typu výrobku:
 - EN 12050-1 alebo EN 12050-2.
- Typ, číslo výrobnej dávky alebo sériové číslo, alebo akýkoľvek iný prvok umožňujúci identifikáciu stavebného výrobku, ako sa vyžaduje podľa článku 11 ods. 4:
 - PUST/PS.R Čerpacie stanice s označením EN 12050-1 alebo EN 12050-2 na typovom štítku.
- Zamýšľané použitie stavebného výrobku, ktoré uvádza výrobca, v súlade s uplatniteľnou harmonizovanou technickou špecifikáciou:
 - Čerpacie stanice určené na čerpanie splaškov s obsahom fekálií s označením EN 12050-1 na typovom štítku.
 - Čerpacie stanice určené na čerpanie splaškov bez obsahu fekálií s označením EN 12050-2 na typovom štítku.
- Názov, registrovaný obchodný názov alebo registrovaná obchodná značka a kontaktná adresa výrobcu podľa požiadaviek článku 11, ods. 5:
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Dánsko.
- NEVZŤAHUJE SA.
- Systém alebo systémy posudzovania a overovania nemennosti parametrov stavebného výrobku podľa ustanovení prílohy V:
 - Systém 3.
- V prípade vyhlásenia o parametroch týkajúceho sa stavebného výrobku, na ktorý sa vzťahuje harmonizovaná norma:
 - TÜV Rheinland LGA Products GmbH, identifikačné číslo: 0197.
Vykonal skúšku podľa EN 12050-1 alebo EN 12050-2 v systéme 3. (popis úloh tretej strany, ako sa uvádzajú v prílohe V)
 - Číslo certifikátu: Certifikát LGA č. 7310150. Typovo skúšaný a monitorovaný. Protokol o skúške LGA č 7313401-01a.
- NEVZŤAHUJE SA.
- Deklarované parametre: Výrobky, na ktoré sa vzťahuje toto vyhlásenie o parametroch, vyhovujú podstatnými vlastnosťami a parametrami nasledovne:
 - Použité normy: EN 12050-1:2001 alebo EN 12050-2:2000.
- Parametre výrobku uvedené v bodoch 1 a 2 sú v zhode s deklarovateľnými parametrami v bode 9.

SE: PrestandadeklARATION

EG prestandadeklARATION enligt bilaga III till förordning (EU) nr 305/2011 (byggproduktförordningen)

- Produkttypens unika identifikationskod:
 - EN 12050-1 eller EN 12050-2.
- Typ-, parti- eller serienummer eller någon annan beteckning som möjliggör identifiering av byggprodukter i enlighet med artikel 11.4:
 - PUST/PS.R Pumpstationer märkta med EN 12050-1 eller EN 12050-2 på typskylten.
- Byggproduktens avsedda användning eller användningar i enlighet med den tillämpliga, harmoniserade tekniska specifikationen, såsom förutsett av tillverkaren:
 - Pumpstationer för pumpning av avloppsvatten innehållande fekalier märkta med EN 12050-1 på typskylten.
 - Pumpstationer för pumpning av fekaliefritt avloppsvatten märkta med EN 12050-2 på typskylten.
- Tillverkarens namn, registrerade företagsnamn eller registrerade varumärke samt kontaktadress enligt vad som krävs i artikel 11.5:
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro
Danmark.
- EJ TILLÄMPLIGT.
- Systemet eller systemen för bedömning och fortlöpande kontroll av byggproduktens prestanda enligt bilaga V:
 - System 3.
- För det fall att prestandadeklARATIONEN avser en byggprodukt som omfattas av en harmoniserad standard:
 - TÜV Rheinland LGA Products GmbH, identifikationsnummer: 0197.
Utförde provning enligt EN 12050-1 eller EN 12050-2 under system 3. (beskrivning av tredje parts uppgifter såsom de anges i bilaga V)
 - Certifikat nummer: LGA-certifikat nr 7310150. Typprobad och övervakad. LGA-testprotokoll nr 7313401-01a.
- EJ TILLÄMPLIGT.
- Angiven prestanda: Produkterna som omfattas av denna prestandadeklARATION överensstämmer med de väsentliga egenskaperna och prestandakraven i följande:
 - Tillämpade standarder: EN 12050-1:2001 eller EN 12050-2:2000.
- Prestandan för den produkt som anges i punkterna 1 och 2 överensstämmer med den prestanda som anges i punkt 9.

TR: Performans beyanı

305/2011 sayılı AB Yönetmeliği Ek III'e uygun olarak performans beyanı
(İnşaat Ürünü Yönetmeliği)

1. Ürün tipi özel tanımlama kodu:
 - EN 12050-1 veya EN 12050-2.
2. Gereken şekil inşaat ürününün Madde 11(4)'e göre tanımlanmasına izin veren tip, parti, seri numarası veya başka bir öge:
 - Etiketinde EN 12050-1 veya EN 12050-2 işareti bulunan PS.R pompa istasyonları.
3. Üretici tarafından öngörülen biçimde ilgili uyumlu teknik özelliklere uygun olarak inşaat ürününün amaçlanan kullanımı ve kullanımları:
 - Dışkı içeren atık suların pompalanmasına yönelik, etiketinde EN 12050-1 bilgisi bulunan pompa istasyonları.
 - Dışkı içermeyen atık suların pompalanmasına yönelik, etiketinde EN 12050-2 bilgisi bulunan pompa istasyonları.
4. Madde 11(5)'e göre gereken şekilde üreticinin adı, tescilli ticari adı veya tescilli ticari markası ve iletişim adresi:
 - Grundfos Holding A/S
Poul Due Jensens Vej 7
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Danimarka.
5. İLGİLİ DEĞİL.
6. Ek V'te belirtilen şekilde inşaat ürününün performansının tutarlılığının değerlendirilmesi ve doğrulanmasına yönelik sistem veya sistemler:
 - Sistem 3.
7. Uyumlu bir standart kapsamındaki bir inşaat ürünüyle ilgili performans beyanı durumunda:
 - TÜV Rheinland LGA Products GmbH, tanımlama numarası: 0197.
EN 12050-1 veya EN 12050-2 göre sistem 3 altında gerçekleştirilen test. (Ek V'te belirtilen şekilde üçüncü taraf işlemlerin açıklaması)
 - Sertifika numarası: LGA Sertifika No. 7310150. Tip test edilmiş ve izlenmiştir. LGA test raporu No. 7313401-01a.
8. İLGİLİ DEĞİL.
9. Beyan edilen performans: Bu performans beyanı kapsamına giren ürünler, aşağıda belirtilen şekilde temel özelliklere ve performans gereksinimlerine uygundur:
 - Kullanılan standartlar: EN 12050-1:2001 veya EN 12050-2:2000.
10. 1. ve 2. noktalarda belirtilen ürünün performansı, 9. noktada beyan edilen performansa uygundur.

Signature

EU declaration of performance reference number: 96235218
Bjerringbro, 15th May 2013



Svend Aage Kaae
Director
Grundfos Holding A/S
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CLONED-56BB0182-B676-4908-8AD9-982069A0BCDB

RUS

PS.R, PS.M**Руководство по эксплуатации**

Руководство по эксплуатации на данное изделие является составным и включает в себя несколько частей:

Часть 1: настоящее «Руководство по эксплуатации».

Часть 2: электронная часть «Паспорт. Руководство по монтажу и эксплуатации» размещенная на сайте компании Грундфос. Перейдите по ссылке, указанной в конце документа.

Часть 3: информация о сроке изготовления, размещенная на фирменной табличке изделия.

Сведения о сертификации:

Насосы типа PS.R, PS.M сертифицированы на соответствие требованиям Технических регламентов Таможенного союза: ТР ТС 004/2011 «О безопасности низковольтного оборудования»; ТР ТС 010/2011 «О безопасности машин и оборудования»; ТР ТС 020/2011 «Электромагнитная совместимость технических средств».

KAZ

PS.R, PS.M**Пайдалану бойынша нұсқаулық**

Атаулы өнімге арналған пайдалану бойынша нұсқаулық құрамалы болып келеді және келесі бөлімдерден тұрады:

1 бөлім: атаулы «Пайдалану бойынша нұсқаулық»

2 бөлім: Грундфос компаниясының сайтында орналасқан электронды бөлім «Төлқұжат, Құрастыру және пайдалану бойынша нұсқаулық». Құжат соңында көрсетілген сілтеме арқылы өтіңіз.

3 бөлім: өнімнің фирмалық тақташасында орналасқан шығарылған уақыты жөніндегі мәлімет

Сертификаттау туралы ақпарат:

PS.R, PS.M типті сорғылары «Төмен вольтты жабдықтардың қауіпсіздігі туралы» (ТР ТС 004/2011), «Машиналар және жабдықтар қауіпсіздігі туралы» (ТР ТС 010/2011) «Техникалық заттардың электрлі магниттік сәйкестілігі» (ТР ТС 020/2011) Кеден Одағының техникалық регламенттерінің талаптарына сәйкес сертификатталды.

KG

PS.R, PS.M**Пайдалануу боюнча колдонмо**

Аталган жабдууну пайдалануу боюнча колдонмо курамдык жана өзүнө бир нече бөлүкчөнү камтыйт:

1-Бөлүк: «Пайдалануу боюнча колдонмо»

2-Бөлүк: «Паспорт. Пайдалануу жана монтаж боюнча колдонмо» электрондук бөлүгү Грундфос компаниянын сайтында жайгашкан. Документтин аягында көрсөтүлгөн шилтемеге кайрылыңыз.

3-Бөлүк: жабдуунун фирмалык тактасында жайгашкан даярдоо мөөнөтү тууралуу маалымат.

Шайкештик жөнүндө декларация

PS.R, PS.M түрүндөгү соргучтар Бажы Биримдиктин Техникалык регламенттин талаптарына ылайыктуу тастыкталган: ТР ТБ 004/2011 «Төмөн вольттук жабдуунун коопсуздугу жөнүндө»; ТР ТБ 010/2011 «Жабдуу жана машиналардын коопсуздугу жөнүндө»; ТР ТБ 020/2011 «Техникалык каражаттардын электромагниттик шайкештиги».

ARM

PS.R, PS.M

Շահագործման ձեռնարկ

Տվյալ սարքավորման շահագործման ձեռնարկը բաղկացած է մի քանի մասերից.

Մաս 1. սույն «Շահագործման ձեռնարկ»:

Մաս 2. էլեկտրոնային մաս. այն է՝ «Անձնագիր: Մոնտաժման և

շահագործման ձեռնարկ» տեղադրված «Գրունդֆոս». Անցեք փաստաթղթի վերջում նշված հղումով.

Մաս 3. տեղեկություն արտադրման ամսաթվի վերաբերյալ՝ նշված սարքավորման պիտակի վրա:

Տեղեկություններ հավաստագրման մասին՝

PS.R, PS.M տիպի պոմպերը սերտիֆիկացված են համաձայն Մաքսային Միության տեխնիկական կանոնակարգի պահանջների՝ TP TC 004/2011 «Ցածրավոլտ սարքավորումների վերաբերյալ», TP TC 010/2011 «Մեքենաների և սարքավորումների անվտանգության վերաբերյալ» ; TP TC 020/2011

«Տեխնիկական միջոցների էլեկտրամագնիսական համատեղելիության վերաբերյալ»:



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ECM: 1261858

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